National University
of Computer & Emerging Sciences

MT1008-Multivariable Calculus

Program: BS(CS)	Semester: Fall 2024
Credit Hours: 03	Pre-requisite: MT1003
Instructor: Dr. Askar Ali	Email: askar.ali@nu.edu.pk

Course Description:
The main objective of the course is to make students proficient in techniques and skills of multivariable calculus. The emphasis will be on application of learned techniques to the solution of problems of science and engineering.

Text & Reference Books:

Text: "Calculus: Early Transcendentals" by James Stewart, 9th Edition Brooks / Cole USA.
Reference: "Thomas' Calculus" by George B. Thomas, Jr., Maurice D. Weir, Joel R. Hass. 15th Edition, Pearson, USA.

Assessment Plan:

Quizzes	10%
Homework assignments	10%
Sessional I	15%
Sessional II	15%
Final exam	50%

Course Learning Outcomes:

Upon successful completion of this course, students will be able to:

1. Comprehend partial derivatives, extremas and chain rule for multivariable functions.
2. Evaluate multiple integrals and discuss their applications.
3. Solve problems involving line integral and surface integral.

Contents:

Parametrization of well-known curves, Line integral of $f(x,y)$ over a plane curve C with respect to s, x and y respectively, line integral of $f(x,y,z)$ over a space curve C with respect to s, x, y and z respectively, line integrals of vector fields, Work	16.2
The fundamental theorem for line integrals, work done, conservative vector fields and potential functions	16.3
Simple curve, closed curve, simple closed curve, connected region, simply connected region, positive orientation, negative orientation, Green's Theorem	16.4
Curl and divergence of a vector field, theorem for checking whether a field is conservative ? irrotational vector field, incompressible vector field	16.5
Parametric surfaces and their areas, Surface integrals.	16.6, 16.7
Stokes' Theorem, The Divergence Theorem	16.8, 16.9